

An End-to-End Framework for Electrical Metrology Digitalization: API-based DCC Integration, XSLT Visualization, and Workflow Automation

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Digital transformation in metrology is evolving from simple document dematerialization toward the creation of truly machine-readable Digital Calibration Certificates (DCCs). This paper presents recent advancements at the National Institute of Metrology, Quality and Technology (Inmetro) in implementing a robust ecosystem for the generation, visualization, and processing of DCCs within electrical metrology laboratories.

The methodology focuses on the development of a Python-based Application Programming Interface (API) that streamlines the creation of DCCs in XML format, in full compliance with the schema proposed by PTB. One of the primary challenges addressed is the inherent complexity of the XML schema for laboratories accustomed to analog processes. To bridge this gap, a simplified JSON structure was developed to map directly to the data fields of conventional certificates. This allows for data input via both JSON and Excel spreadsheets, making the process accessible to laboratory staff with basic IT skills while maintaining technical rigor.

A key contribution of this framework is the development of a custom DCC viewer using Extensible Stylesheet Language Transformations (XSLT). This tool addresses the need for human readability by rendering the complex XML data into a visual format that closely mirrors the official calibration certificate templates used by Inmetro. This ensures that, while the data is machine-optimized, it remains intuitive and verifiable for human end-users and quality managers, facilitating the transition from paper-based systems.

Furthermore, the study details the practical implementation of embedding the DCC XML within a PDF/A-3 file, providing a hybrid solution that combines familiar digital signatures with machine-readability. The effectiveness of the framework was validated through a pilot project involving Inmetro's LAMPE and LACEL laboratories. By utilizing tools such as NI LabVIEW to extract data from DCCs and automatically update reference databases, a fully automated calibration workflow was demonstrated. The results indicate that the use of DCCs eliminates manual transcription errors and significantly reduces processing time for certificates with extensive data points.

The study concludes that the combination of modular APIs and standardized visualization tools is essential for global interoperability. This framework serves as a scalable model for National Metrology Institutes transitioning into the Metrology 4.0 era, ensuring that digital certificates

are both technologically advanced and practically accessible.

References:

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